Reactions of Payoff phase:

Step 6: Glyceraldehyde-3-phosphate is oxidized by NAD^+ which gets reduced to NADH and H^+. This reaction is exergonic, releasing energy which is utilised to phosphorylate the Glyceraldehyde-3-phosphate to form high energy molecule 1,3-bisphosphoglycerate. Here the inorganic phosphate (P_i) is used for phosphorylation. Glyceraldehyde-3-phosphate dehydrogenase catalyses this step.

Step 7: The 1,3-bisphosphoglycerate is converted into 3-phosphoglycerate transfering one of its phosphate group to ADP to form ATP. The enzyme involved in this process is phosphoglycero kinase

Step 8: 3-Phosphoglycerate is isomerised to 2-phosphoglycerate by the enzyme phosphoglycerate mutase

Step 9: 2-Phosphoglycerate loses a molecule of water, becoming phosphoenolpyruvate (PEP). This step is catalysed by enolase enzyme. PEP is an unstable high energy molecule, poised to lose its phosphate group in the final step of glycolysis.

Step 10: In the final step, PEP readily loses its phosphate group to ADP and forms pyruvate, the end product of glycolysis. In this reaction one molecule of ATP is formed. This reaction is catalysed by pyruvate kinase (Figure 3.4).

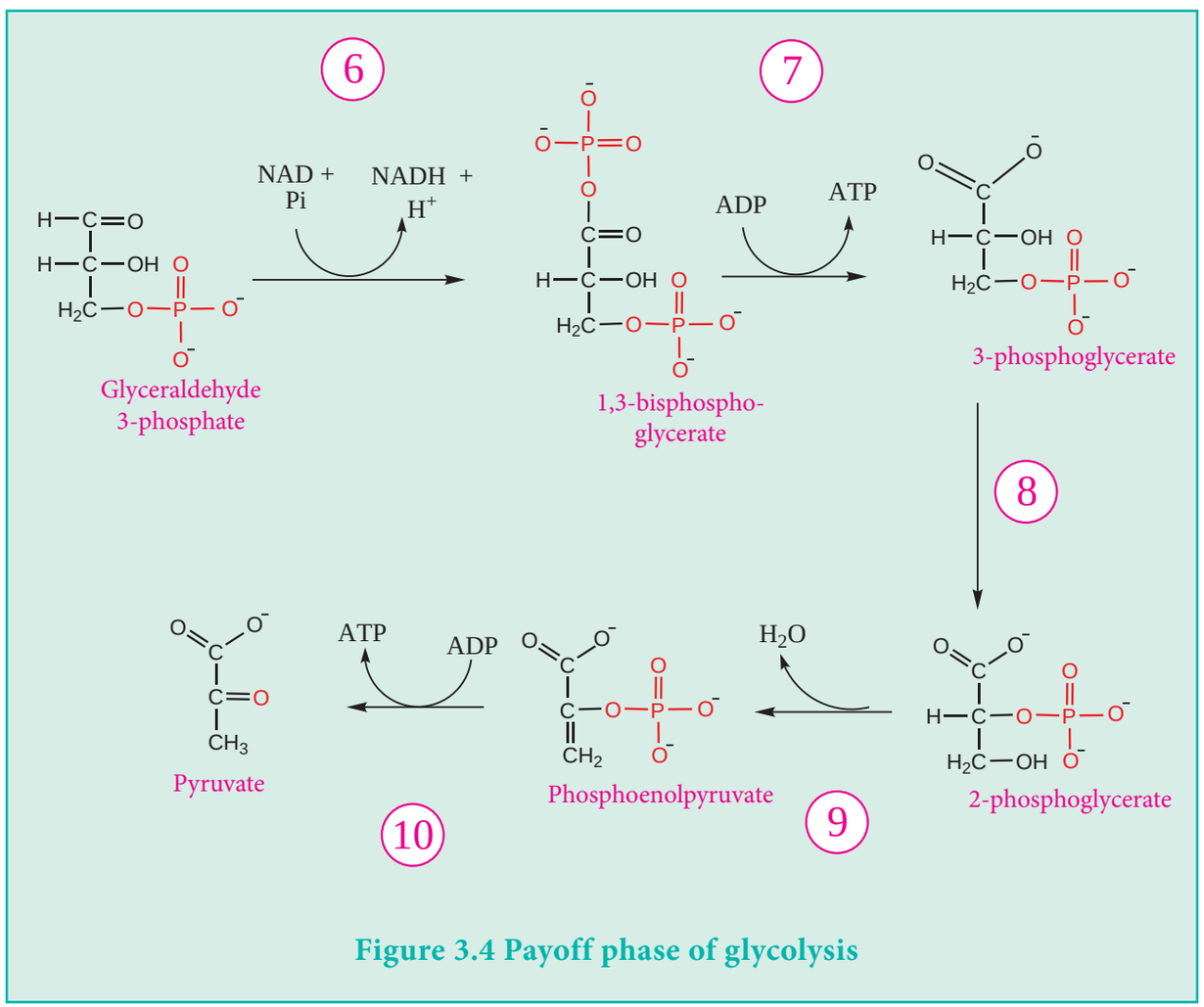


Figure 3.4 Payoff phase of glycolysis

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